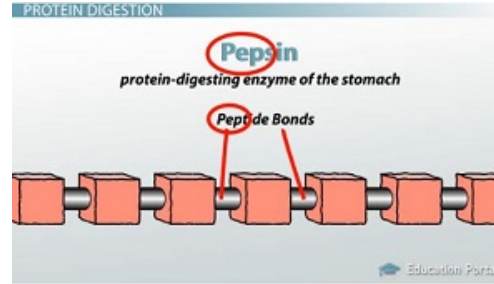


Enzymes:

Many enzymes are proteins (NOT ALL, RNA can also acts as enzyme)



RECAP (Lecture 6)

Enzymes

Simple Enzymes:
That consist only protein

Holoenzymes:

Apo enzyme (protein)

cofactor (non-protein)

Metal ions
(INORGANIC)

Example: Alcohol dehydrogenase (Zn^{2+})

Conenzyme
(ORGANIC)

Prosthetic
➤ TIGHTLY BOUND
➤ REMAIN UNCHANGED

Example: Heme

Cosubstrate
➤ LOOSELY BOUND
➤ ALTERED DURING REACTION

**Example: NAD⁺
FAD, Coenzyme A**

Vitamins



REMINDER....

RECAP
(Lecture 6)

Red-ox



Coenzyme

Reduced Coenzyme

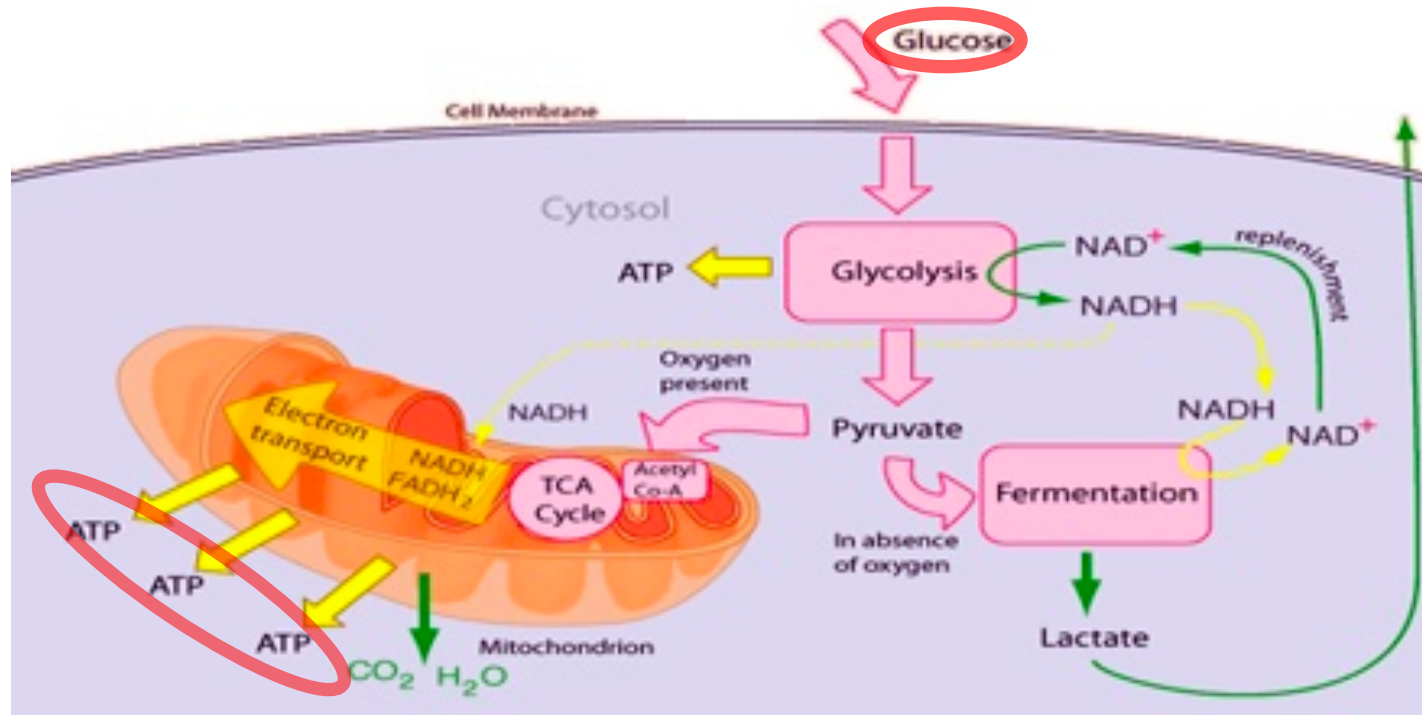


Acetyl group



Acyl transporter

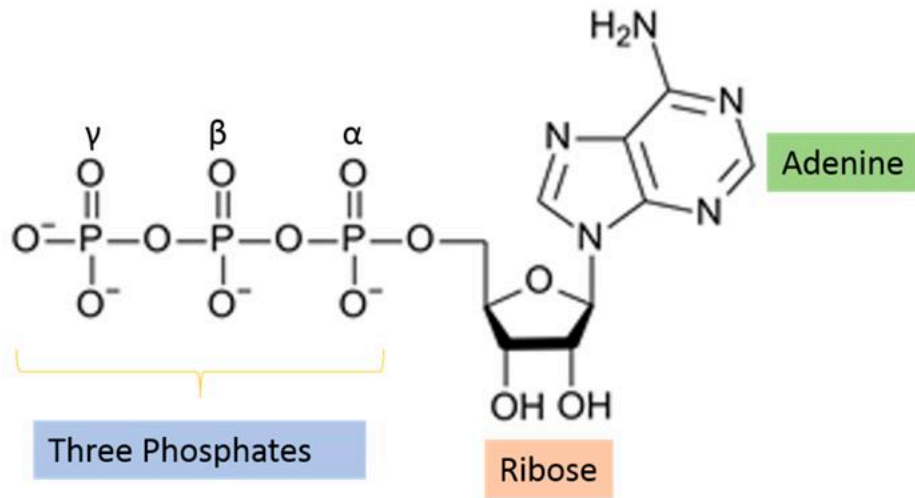
Glucose Metabolism (Glucose $\rightarrow$ ATP)



<https://www.assignmentpoint.com/science/health/describe-about-glucose-metabolism.html>

Dr. P. Satpati,
BSBE, IIT Guwahati₃

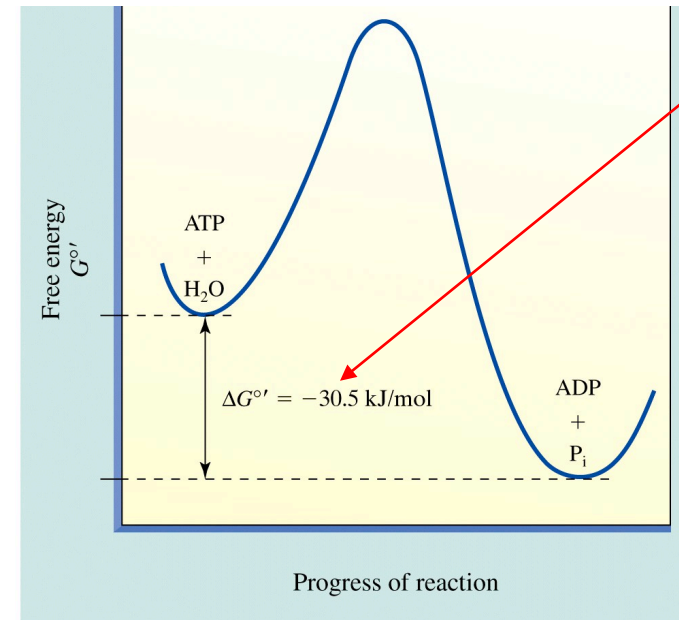
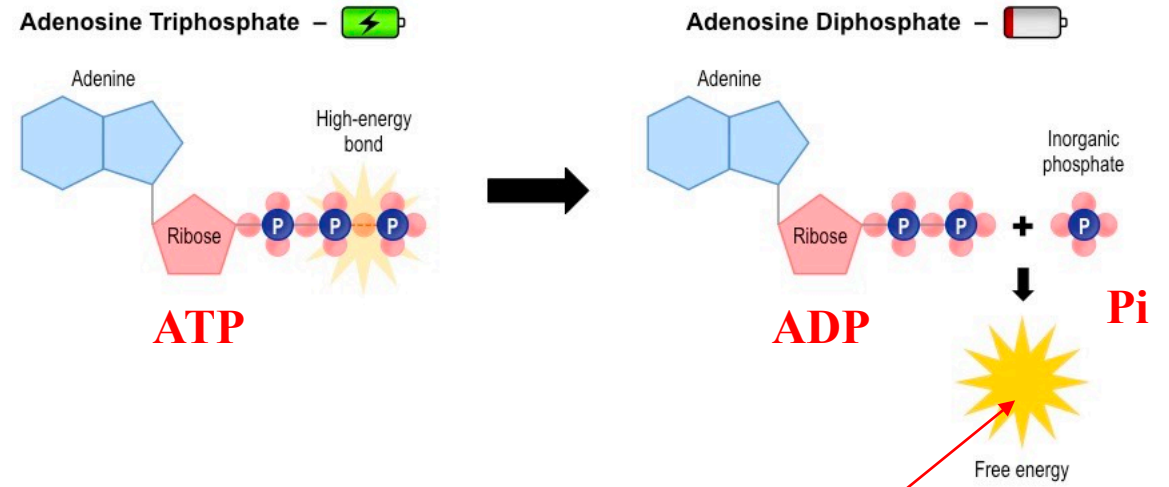
What is ATP ?



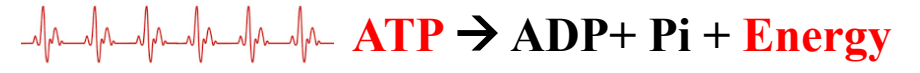
Adenosine triphosphate (ATP)

ATP is uphill. Thus, ATP production from ADP+Pi require energy (from food).

What is the use of ATP?

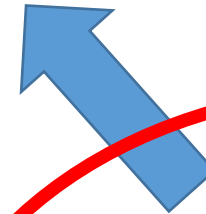
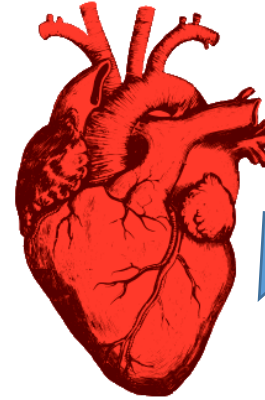
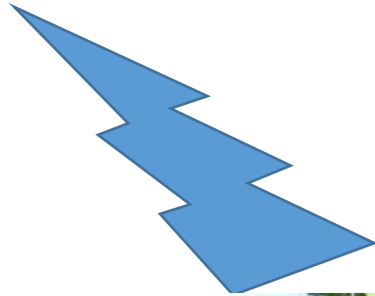


₹ATP



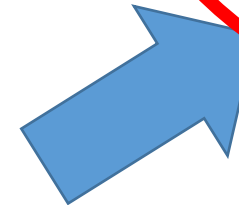
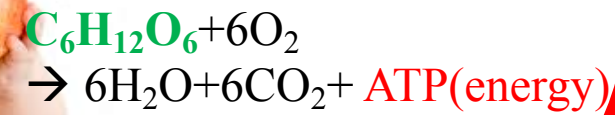
Photosynthesis:

Energy from sun is converted into chemical bonds (Glucose)

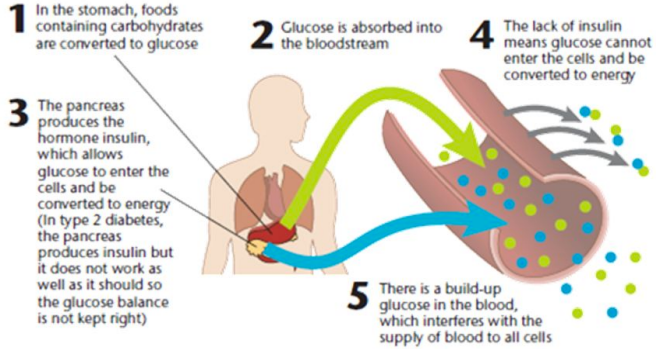


Cellular respiration:

Glucose broken down into usable form of Energy **ATP**.



Carbohydrates>Glucose>Energy



- It is insulin that allows glucose to enter the body's cells
- The absence of insulin results in abnormally high blood sugar and eventually DIABETES.

Carbohydrates

H_2O

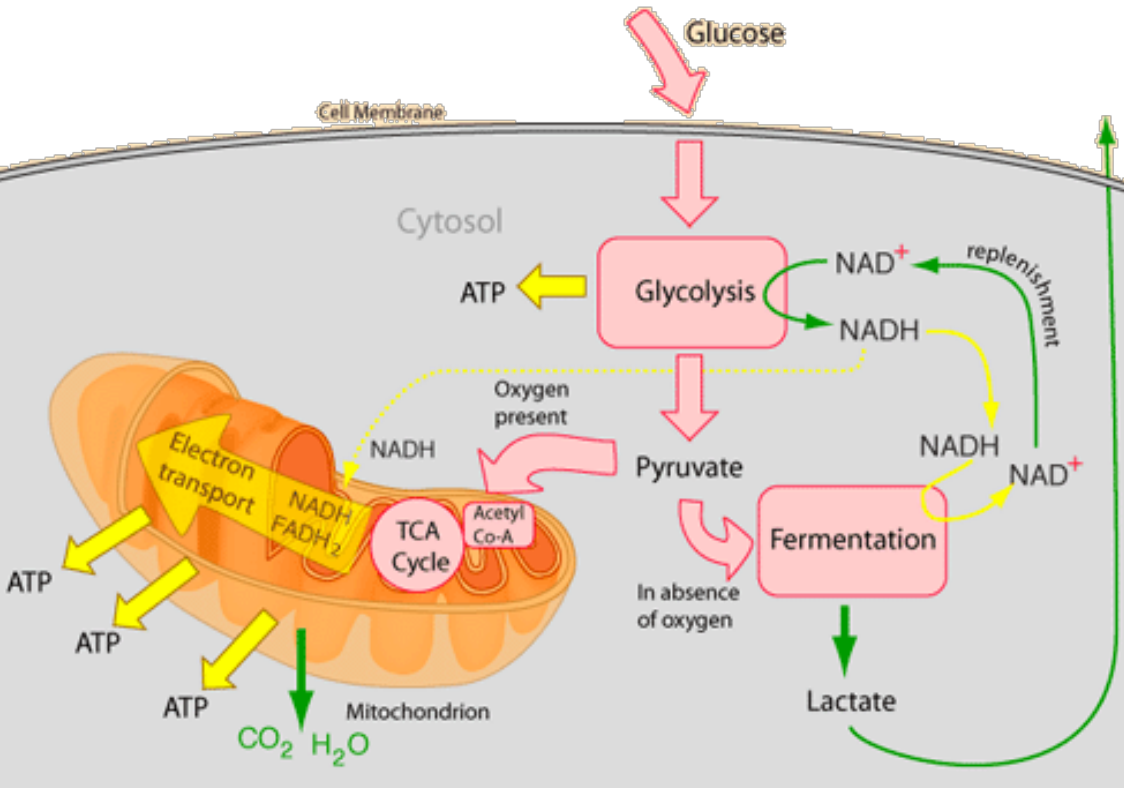
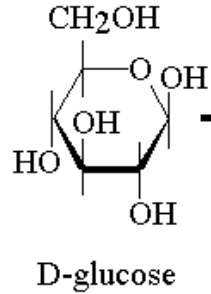
Monosacharides/Glucose

Stage 1: Hydrolysis

Carbohydrates>Glucose>Energy

- 1 In the stomach, foods containing carbohydrates are converted to glucose
- 2 Glucose is absorbed into the bloodstream
- 3 The pancreas produces the hormone insulin, which allows glucose to enter the cells and be converted to energy (in type 2 diabetes, the pancreas produces insulin but it does not work as well as it should so the glucose balance is not kept right)
- 4 The lack of insulin means glucose cannot enter the cells and be converted to energy
- 5 There is a build-up glucose in the blood, which interferes with the supply of blood to all cells

- It is insulin that allows glucose to enter the body's cells
- The absence of insulin results in abnormally high blood sugar and eventually **DIABETES**.

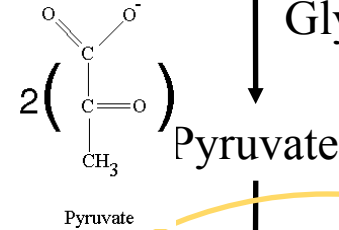


Carbohydrates

H₂O

Monosacharides/Glucose

Glycolysis



Acetyl CoA

Citric Acid Cycle/TCA

Mitochondrion

Electrons/Reduced coenzyme

Electron Transport Chain

Oxidative phosphorylation

ATP

ADP+Pi

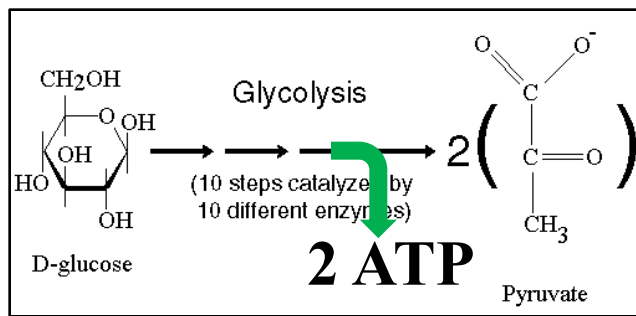
Stage1: Hydrolysis

Stage2: Acetyl CoA production

Stage2: Energy production



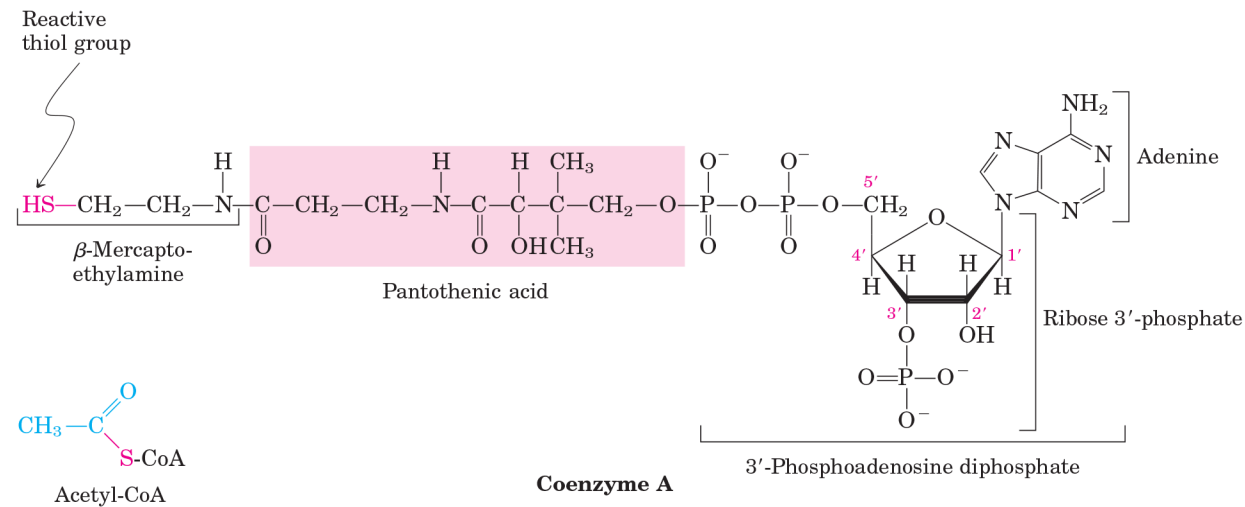
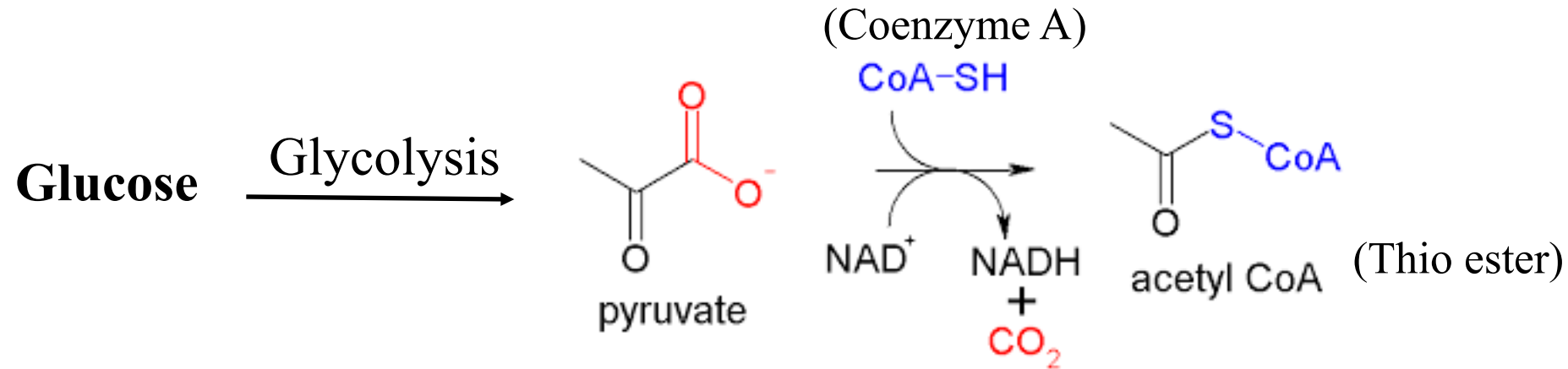
What happens to the food we eat ?



Before the citric acid cycle



RBC, Mitochondria

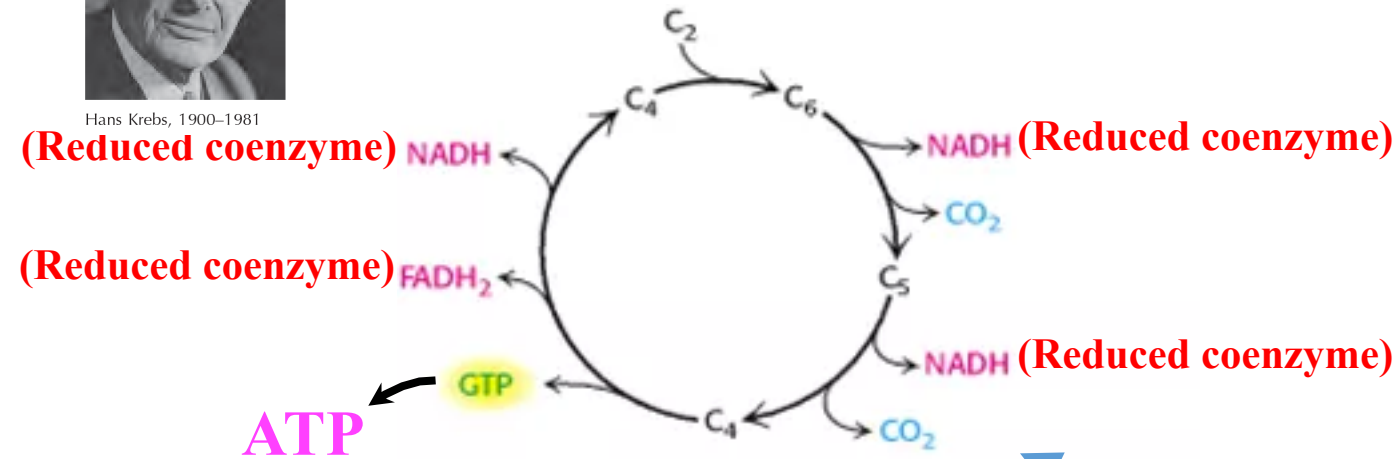


Taken from: Principles of Biochemistry- Albert Leningher



Hans Krebs, 1900–1981

Citric acid cycle (CAC)/ Kreb's cycle



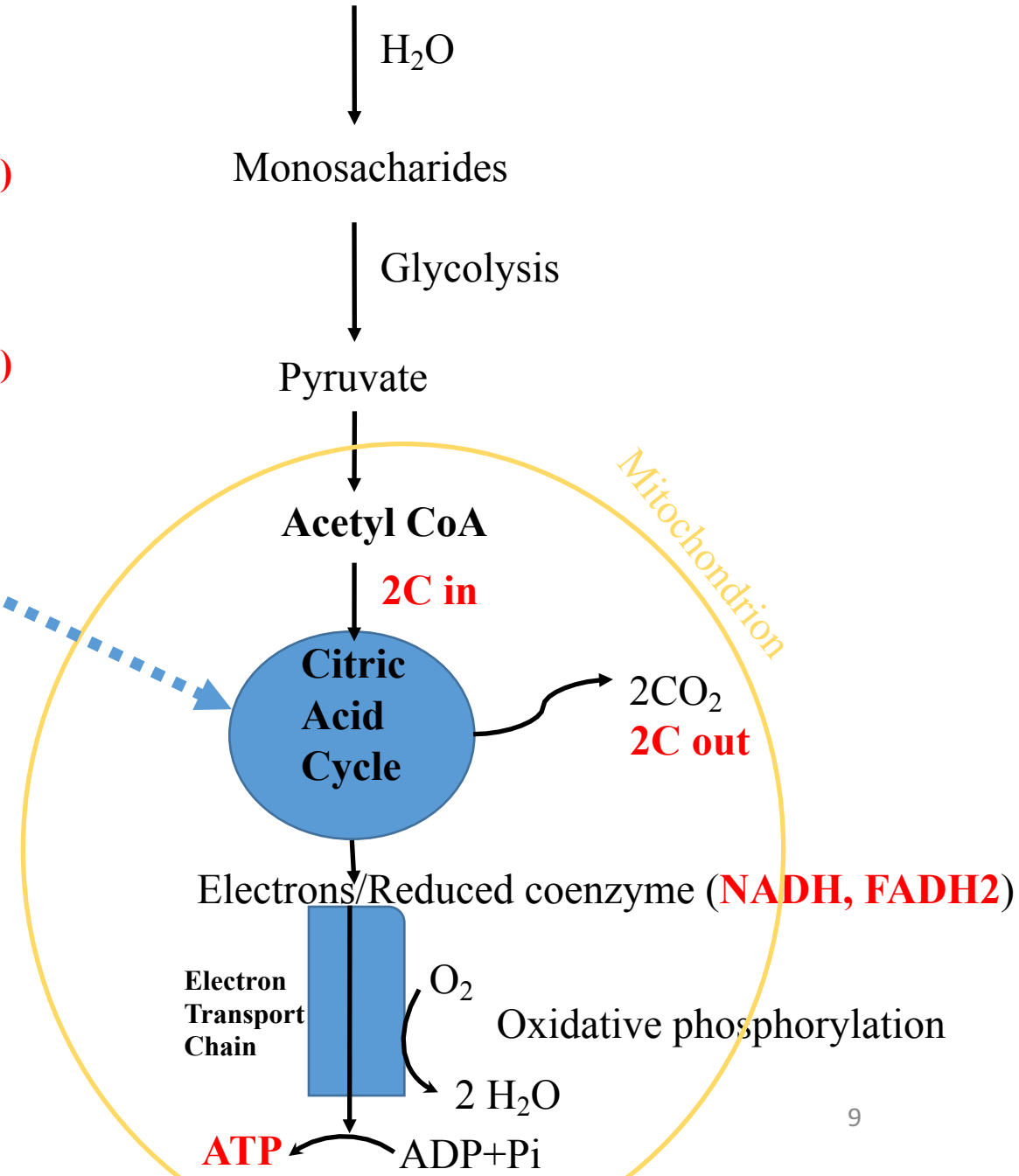
Summary :

1. 2C in and 2C out
2. Reduced coenzyme formation

3. ₹

- ❖ ATP
- ❖ NADH
- ❖ FADH₂

Carbohydrates



Citric acid cycle/ Kreb's cycle – Details

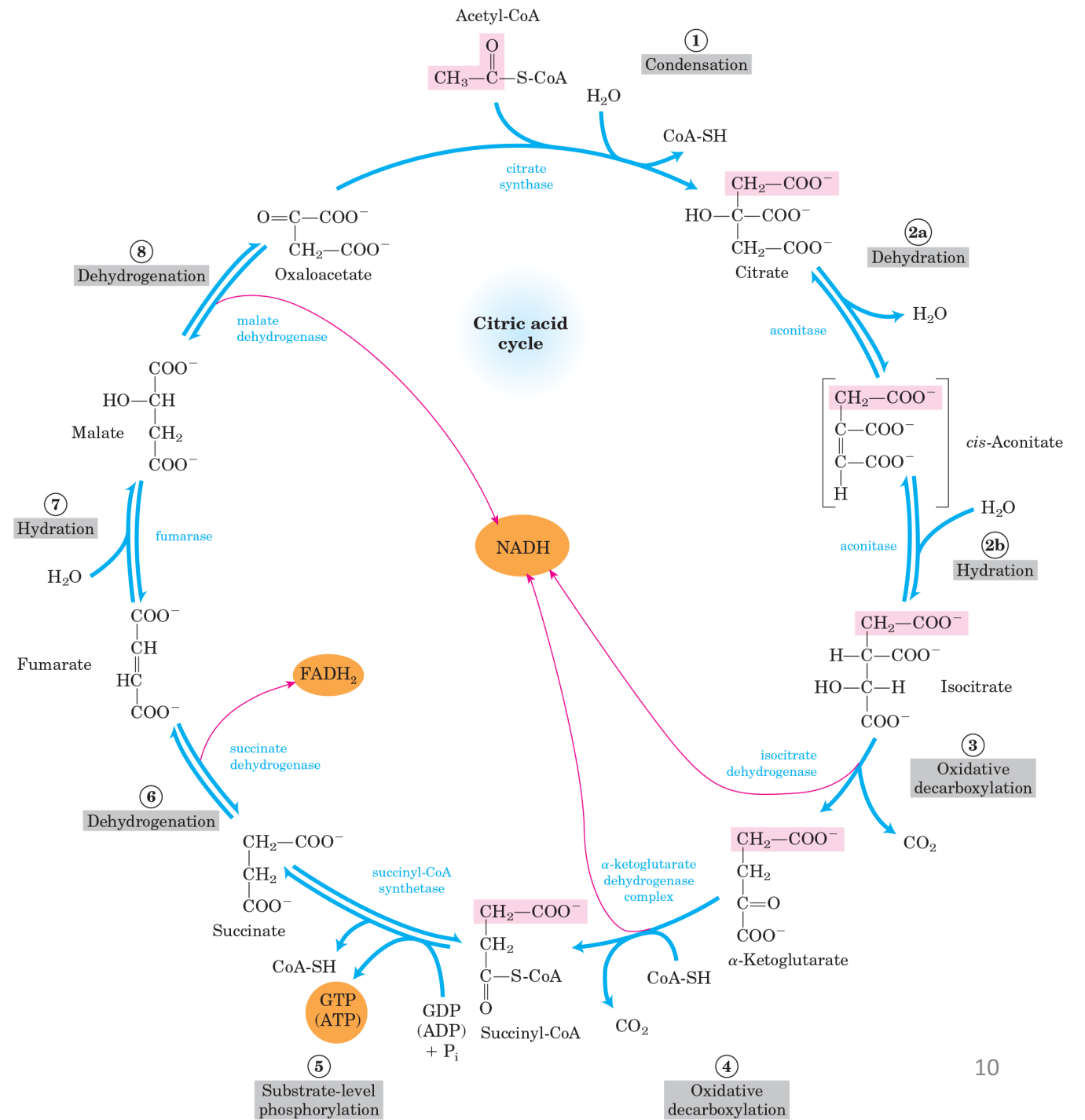


Points to be noted:

(1) 2C out from CAC are different from Acetyl carbon.

(2) 2° alcohol is easily oxidized.
Step 8 is facilitated by (2a,2b)

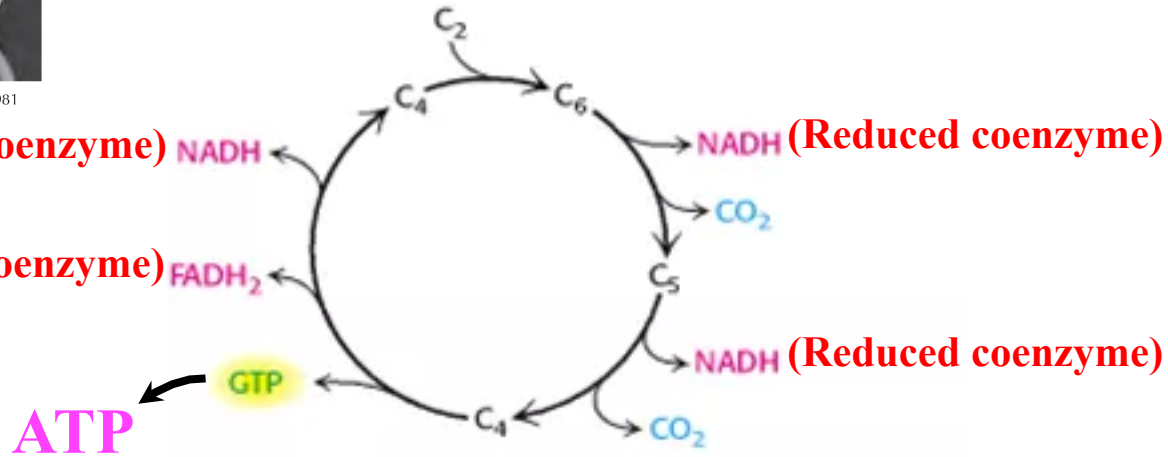
(3) Cyclic (produces oxaloacetate at the end)



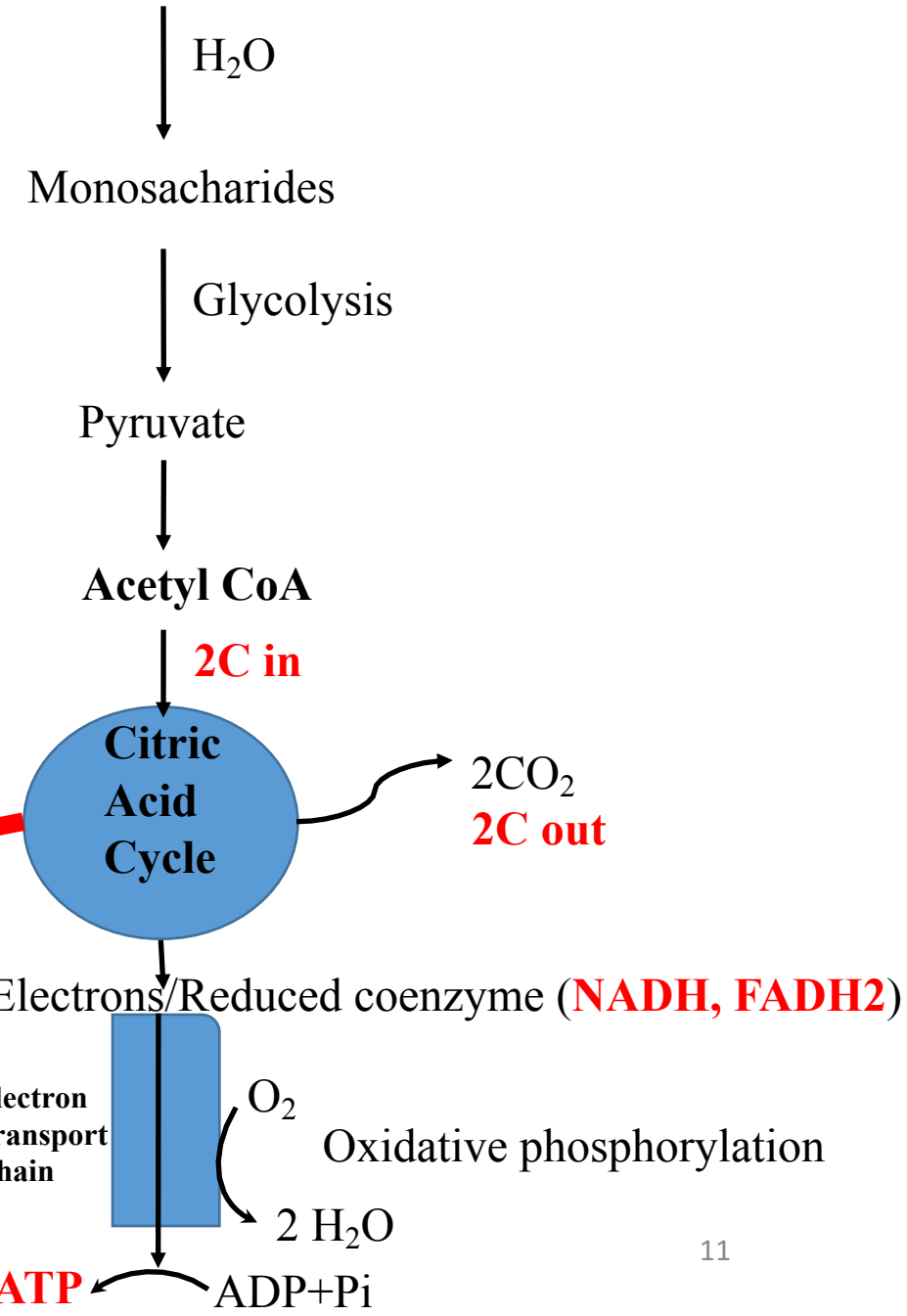


Hans Krebs, 1900–1981

Citric acid cycle (CAC)/ Kreb's cycle



Carbohydrates



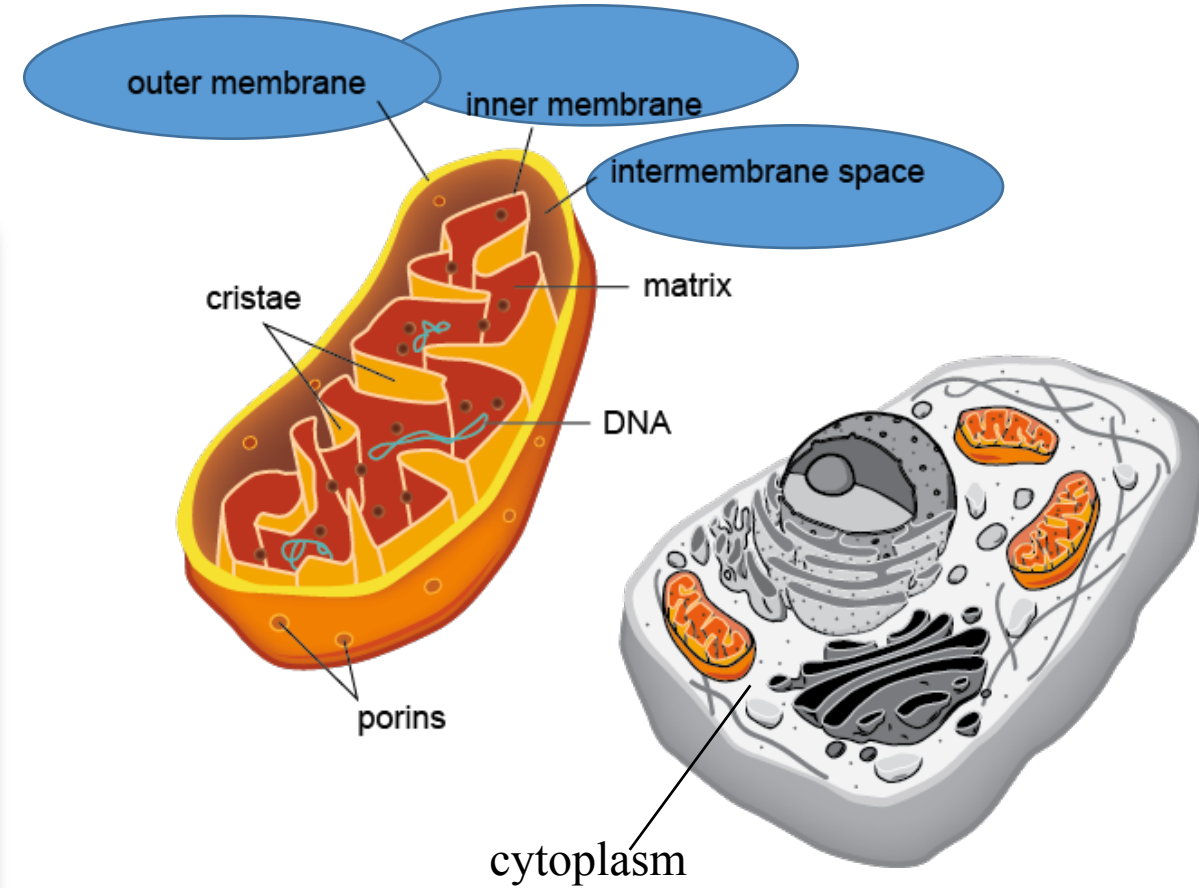
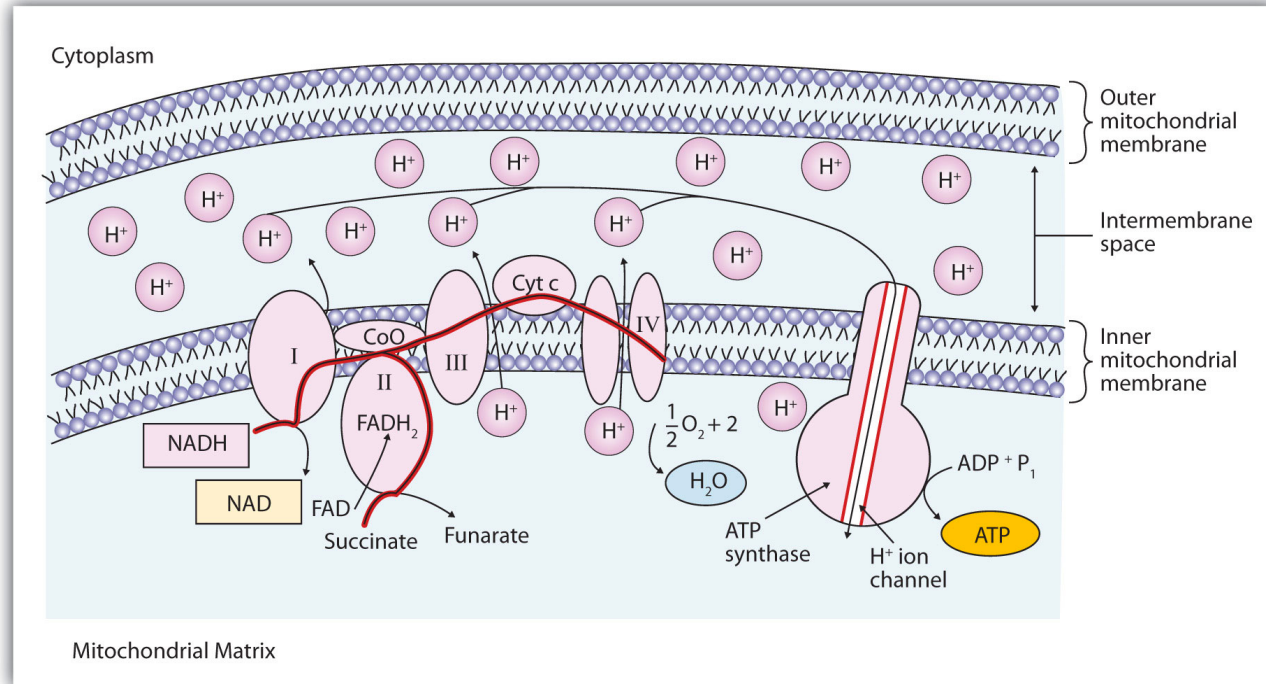
Summary :

1. 2C in and 2C out
2. Reduced coenzyme formation

3. ₹

- ❖ ATP
- ❖ NADH
- ❖ FADH₂

Where does the Electron transport chain (ETC) and ATP production takes place ?



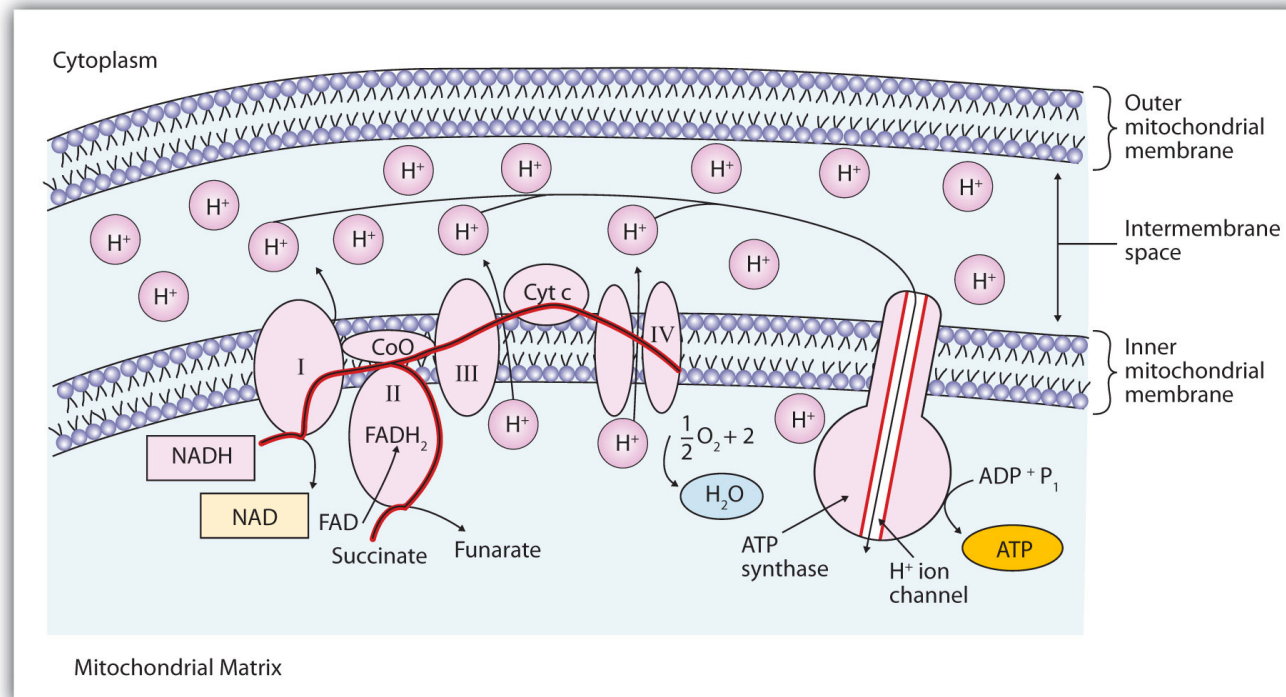
(ETC) = Complex I - IV

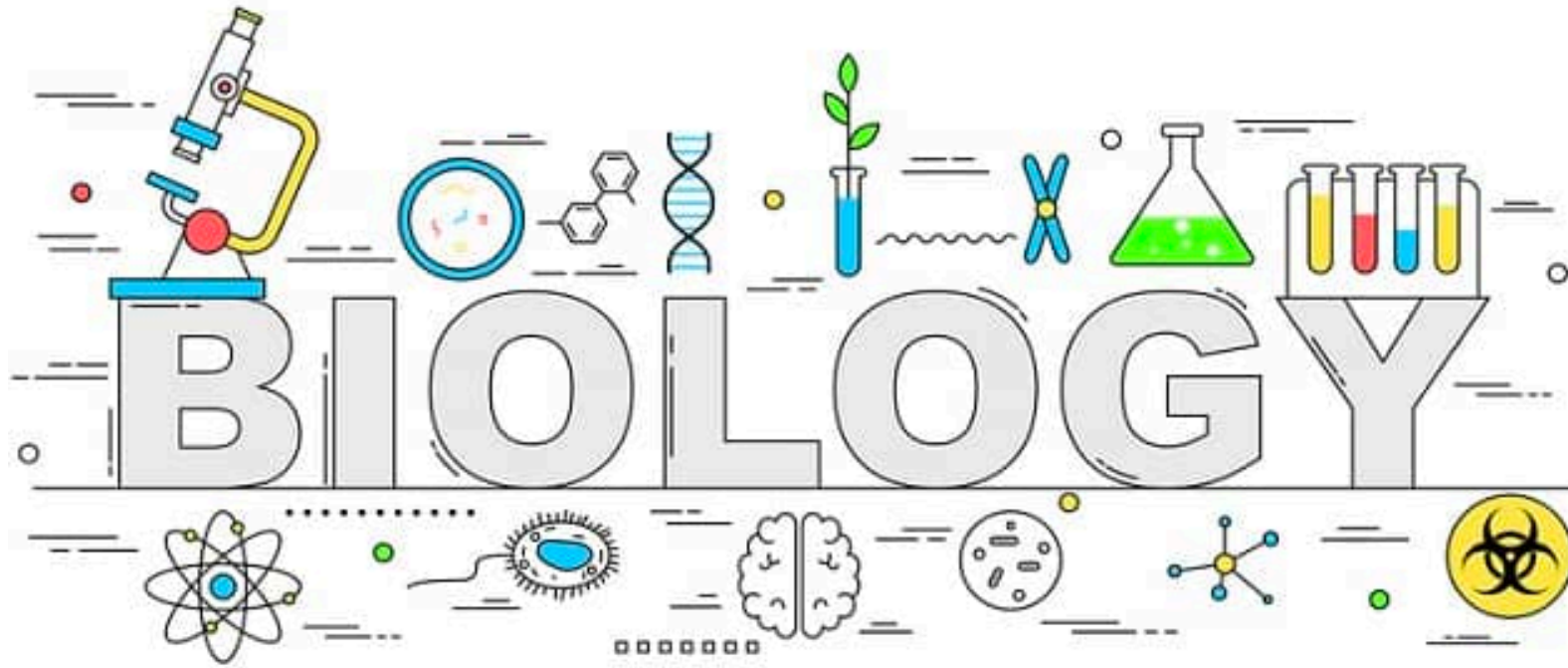
ATP synthase
(Complex V)

ATP

Summary

1. CAC produces **reduced enzymes (NADH, FADH₂)** + ATP
2. NADH, FADH₂ enters ETC. Pumps H⁺ from matrix to inter membrane space of mitochondria.
3. Accumulation of H⁺ in the inter membrane of mitochondria. Chemical and electrical potential energy.
4. Diffusion of H⁺ through ATPase from inter to inner membrane → synthesis of ATP





Next: Glucose Metabolism (cont..)